AP CALCULUS AB

Unit 1 • Lesson 1-2 • Notes (filled in)

Lesson 1-2 • Limits to a Point of a Function

Notes (filled in)

Learning goals

- Evaluate limits by direct substitution and with the limit properties.
- Rewrite $\frac{0}{0}$ forms by factoring, using conjugates, and combining fractions.
- Handle $\frac{k}{0}$ forms, piecewise and absolute value functions, and the special trigonometric limits.

Vocabulary. direct substitution, indeterminate form, conjugate, complex fraction, squeeze theorem.

1. Start with direct substitution

Key idea

Polynomials, rational functions (where defined), roots, sin, cos, e^x and $\ln x$ are continuous on their domains, so $\lim_{x\to c} f(x) =$ $f(c)$ for c in the domain.

Properties: limits of sums, differences, products and constant multiples are the sums, differences, products and multiples of the limits; for a quotient, the denominator's limit must be nonzero.

Example 1

Evaluate $\lim_{x\to -1}(x^3 - 4x + 2)$ and $\lim_{x\to \pi/3} 2\cos x$. If $\lim_{x\to 4} f(x) = 3$ and $\lim_{x\to 4} g(x) = -2$, find $\lim_{x\to 4}\left(2f(x) - g(x)^2\right)$.

$-1 + 4 + 2 = 5. \quad 2 \cdot \frac{1}{2} = 1. \quad 2(3) - (-2)^2 = 2.$

2. What substitution tells you

Key idea

a real number	that number is the limit
$\frac{0}{0}$	indeterminate: rewrite (factor, conjugate, combine fractions, trig limit)
$\frac{k}{0}, k \neq 0$	the limit is infinite or does not exist: check the sign on each side

Example 2 • Factoring

$$\lim_{x\to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$$

$$\frac{(x-3)(x+3)}{(x-3)(x+1)} = \frac{x+3}{x+1} \to \frac{6}{4} = \frac{3}{2}.$$

Example 3 • Conjugate

$$\lim_{x\to 0}\frac{\sqrt{x+9}-3}{x}$$

$$\frac{(\sqrt{x+9}-3)(\sqrt{x+9}+3)}{x(\sqrt{x+9}+3)}=\frac{x}{x(\sqrt{x+9}+3)}=\frac{1}{\sqrt{x+9}+3}\to\frac{1}{6}.$$

Example 4 • Complex fraction

$$\lim_{x\to 2}\frac{\frac{1}{x}-\frac{1}{2}}{x-2}$$

$$\frac{\frac{2-x}{2x}}{x-2}=\frac{-1}{2x}\to-\frac{1}{4}.$$

Example 5 • $\frac{k}{0}$ and absolute value

$$\lim_{x\to 2}\frac{x+3}{x-2}\quad\text{and}\quad\lim_{x\to -1}\frac{x^2-1}{|x+1|}$$

First: $\frac{5}{0}$. For $x<2$: $-\infty$; for $x>2$: $+\infty$. The limit does not exist.
Second: for $x<-1$, $|x+1|=-(x+1)$, the quotient is $-(x-1)\to 2$; for $x>-1$ it is $x-1\to -2$. Does not exist.

3. Trigonometric limits and the squeeze theorem

Key idea

$\lim_{x\to 0}\frac{\sin x}{x}=$ __1__ $\lim_{x\to 0}\frac{1-\cos x}{x}=$ __0__

Squeeze theorem: if $g(x)\le f(x)\le h(x)$ near c and $\lim_{x\to c}g=\lim_{x\to c}h=L$, then $\lim_{x\to c}f(x)=$ __L__.

Example 6

$$\lim_{x\to 0}\frac{\sin 5x}{x}\quad\text{and}\quad\lim_{x\to 0}x^2\sin\frac{1}{x}$$

$5\cdot\frac{\sin 5x}{5x}\to 5$. $-x^2\le x^2\sin\frac{1}{x}\le x^2$ and both bounds $\to 0$: the limit is 0.

Common mistakes

- Concluding that a $\frac{0}{0}$ limit is 0, 1 or does not exist without rewriting.
- Cancelling terms instead of factors.

Check your understanding

1. $\lim_{x\to 1}\frac{x^3-1}{x-1}=$ __3__. 2. $\lim_{x\to 0}\frac{\sin 2x}{\sin 5x}=$ __$\frac{2}{5}$__. 3. $\lim_{x\to 3^-}\frac{1}{x-3}=$ __$-\infty$__.